# Travel Demand Model Formulation and Implications to Realize User Behavior in Response to Vehicle Miles Travelled Reduction Strategies

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**Abstract:**

*Vehicle Miles Travelled (VMT) has become a major concern in transportation planning because of large dependency in automobile travel. A number of transportation agencies propose proactive ideas of reducing VMT. However, there are limited number of studies in literature which show promise in quantification of methods to demonstrate how user behavior will react to strategies aimed at VMT reduction. In this paper, we propose a set of models to incorporate decision maker's policies and user behavior in a trip based travel demand model. We propose three strategies (1) a VMT based pricing, (2) gas tax, and (3) commute efficiency to envisage reduction in VMT. First we formulate a base case to simulate the current travel condition, and proposed three models aimed at VMT reduction strategies possible shift of destination, use of alternate mode, or use of different path because of larger cost imposed on the network for private vehicle travel. The set of models constitute destination choice, mode choice and route choice algorithms to realize this effect. Further we consider demand to be elastic (as opposed to static demand in first three models) and apply three strategies to realize the effect of VMT. The destination choice and mode choice are formed as multinomial logit functions and the route choice is considered as frank wolfe based assignment. The proposed model is applied to a multimodal transportation network with large representation of transit. The motivation for using*

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*a multimodal transportation network is to examine the effect of user behavior elasticity in response to impeding strategies for highway travel and potential shift to transit. The results show that each model has its own advantages to some extent. The case study results show that such tool can be used as a tool for VMT reduction (or policies of similar nature) in transportation planning and policy decision making.*

# 1 INTRODUCTION

Vehicle Miles Travelled (VMT) has become a major concern in the transportation planning world with increased automobile dependence. Ever growing VMT create apprehension among transportation decision makers as it leads to increase in congestion, and lowers the level of service of the designed facilities. In addition higher VMT leads to Green House Gas (GHG), greater fuel consumption, increased congestion, larger consumer spending in automobile travel, and increase energy usage (Rentziou, Gkritza, and Souleyrette, 2012). A number of planning agencies around the world are constantly thinking about innovative ways to reduce VMT. For example, the proposed Federal Surface Transportation Policy and Planning Act of 2009 includes goals to "reduce national per capita motor vehicle miles traveled on an annual basis" and "increase the total usage of public transportation; proposals include targets to reduce per capita vehicle travel 16%; California law requires regional governments to develop smart growth transport and land use plans that reduce VMT (CPDR 2008).

The mechanisms developed to reduce VMT vary from supply constraints to pricing incentives, all aimed at producing more efficient travel patterns. However, many of the proposed VMT reduction mechanisms are rarely modeled within an empirically calibrated transportation

demand model or tested to determine how they would perform in a true transportation environment. Further, a limited number of studies show application of travel behavior techniques to VMT reduction policies within a single modeling framework and compare the effectiveness of such policies.

Currently, there are two types of feasible methods, one is a long-term solution to improve public transport such that the mode shift can cause a large VMT reduction and the other is a short-term solution to influence traveler behavior by imposing alternative smart transportation policies so that greater VMT reduction is achieved in the system. Although the former is the ideal scenario, for a long-term sustainable solution, the latter needs more careful analysis before determining additional cost to the user to achieve substantive reductions. In both cases there is a need for the development of travel demand or behavioral models. Such models are capable of examining the travel behavior of road users and their responses to impedances such as policies, to achieve VMT reductions.

In this paper we seek to understand how various policy decisions will enhance VMT reduction based on a robust behavioral model. Next, we develop various models to incorporate strategies to reduce system level VMT for a transportation network. The models developed in this paper are at the macroscopic level and can be used for any multimodal transportation network. In this paper the transit system is integrated with the highway network and is responsive to traffic congestion on the road network. The proposed traffic assignment procedure is a multi-class considering all vehicle classes are assigned to the network simultaneously.

The developed models simultaneously consider planner (or policymaker) and user goals; a planner's sole objective is to design policies envisioning certain goals while users (drivers)

accordingly respond to policy decisions while minimizing their travel-cost/travel-time. Thus, considering a user's reaction to a given policy, an optimal decision is one which favors the criteria of meeting certain goals of the policy maker. The methodology developed in this paper is generic and can be applied to any transportation network and policies similar to reduction of VMT or derivatives of thereof.

The remainder of this paper is organized into five sections. The first section reviews the literature on VMT reduction mechanisms both from the supply side and the travel demand side. The second section presents the modeling framework and policy formulations of the tested VMT reduction mechanisms. The third section describes the case study area and data. Results of the modeling application are presented in the fourth section followed by a discussion, conclusions and suggestions for further research in the fifth section.

# 2 BACKGROUND

The background section is arranged into three parts. First, literature review of research demonstrating why reduction in VMT is critical; second, how these reductions are achieved and third, what are the analytical methods adapted for such reduction.

## 2.1 Why VMT Reduction is Critical

Reducing the number of vehicle miles travelled (VMT) per capita has become a serious policy goal in recent years. VMT reductions are linked to a host of benefits including congestion and emissions reductions, facility cost savings, consumer savings, energy conservation and improved public health (Litman, 2010). The key to substantial congestion reductions has been associated

with efforts that reduce VMT through pricing (Giuliano, 1992). Pricing is also the subject of other studies that studies have found that emissions mitigation benefits coincide with VMT reductions (Sharma and Mishra, 2011). Public health is impacted indirectly by VMT as a result of less exercise and more pollution. Studies have found that there is also a direct link to vehicle miles travelled and traffic fatalities (Martinez-Schnell & Zaidi, 1989).

Beyond the benefits that are gained from VMT reductions, there are regulatory reasons that planners would need to reduce emissions. In 2006 California passed Assembly Bill 32, called the California Global Warming Solutions Act. The act required the equivalent of a 25% reduction in Greenhouse Gases (GHG) by 2020, mandating that emissions in that year be at the 1990 level (Michael Hanemann, 2007). The primary tool to achieve GHG reductions was defined in a later bill, Senate Bill 375, which required a stop to VMT growth and progress towards mileage reductions. Similar to the California bills, Washington State adopted RCW70.235.020, in 2008, which sets state GHG targets. To supplement the bill RCW 47.01.440, was enacted in the same year requiring per capita VMT reductions of 50 percent by the year the next 40 years (Howard, 2010). Many urban regions in the US, Australia, and Europe, for example, are making significant efforts to shift automobile to non-automobile transportation modes that would imply urban land-use patterns to much higher-density living and toward greater use of mass transit there by reducing VMT (Moore, Staley, and Poole, Jr, 2010). These regulations are examples of a growing number of laws that seek to mitigate congestion and climate change, encourage transit usage, and increase public health all through the mechanism of VMT reductions.

## 2.2 How VMT Reduction is Achieved

Once the motivations and benefits are established, the question turns to how reductions might be achieved. The literature generally focuses on demand management as the strongest policy candidate for VMT reductions. Perhaps the most effective both theoretically and empirically is highway pricing. Many studies examine congestion pricing and measure VMT benefits (Daniel & Bekka, 2000). These studies fall under the First-best theory in which pricing is meant to capture both the marginal and average social costs of the externality. Poole (1992) finds likely reductions in demand and thus VMT when a congestion charge is added to a new toll link.

Other pricing approaches fall under the theory of the Second-best which include increases to vehicle or fuel taxation (Fullerton & West, 2002; Sterner, Dahl, & Franzen, 1992), charges based on the number of vehicle miles traveled (Greene, 2011) and charging for travel that occurs within a cordon, such as that case with the cordon in London (Beevers & Carslaw, 2005) and in Singapore (Chin, 1996).

The second category of demand side policies work to encourage more efficient travel by reorganizing urban form (Anderson, Kanaroglou, & Miller, 1996; Camagni, Gibelli, & Rigamonti, 2002; Muniz & Galindo, 2005; Williams et al., 2000) or increasing urban density such that alternative modes are either necessary or more accessible (Chatman, 2003).

Other studies focus on encouraging a balance between residential and job location (Horner, 2002; Scott, Kanaroglou, & Anderson, 1997) by developing models that minimize commute times. Other mechanisms attempt to reduce congestion and VMT with a reduction in the demand for light-duty vehicle (LDV) transport. This is accomplished through a list of measures specified from the 1990 Clean Air Act (US EPA, 1990). The set of tools, called

Transportation Control Measures (TCM) are mostly policy-based options to encourage programs that make more efficient use of LDVs during commute hours, increase transit mode share, promote non-motorized travel, increase vehicle occupancy with ride share programs and replace home-base work travel with telecommuting (Hall, 1995; Zhou et al., 2010). Some studies have looked into the effects of telecommuting and found it alone to have an impact. Choo et al (2005), developed a time series study and found VMT to be affected by telecommuting with 94% confidence, though the impact was found to be relatively small, on the magnitude of .80%. Rentziou, Gkritza, and Souleyrette (2012) found that a series of factors such as increased fuel cost, urban density, availability of transit, socioeconomic and demographic data depends on VMT.

### 2.3 Analytical Methods for VMT Reduction

While there seems to be no shortage of ways to reduce VMT, the literature is limited on how to evaluate these potential policies. Studies in the US are limited to Metropolitan Planning Organization (MPO) scenario studies, which attempt to model the effects of future development patterns based rather that analyze policy formulation (Bartholomew, 2005; Johnston, 2006). These efforts generally do not attempt to measure the effectiveness of different policy options. In the UK, a much more robust set of models has been developed but modeled policies tend to focus on land use polices or transport supply rather than transportation demand management (Webster, Bly, Paulley, & Brotchie, 1988). The SPARTACUS model is perhaps the most comprehensive in its policy analysis, modeling both land-use and transport, the results indicated road pricing was the most effective method of achieving VMT reductions (Lautso & Toivanen, 1999). Rentziou, Gkritza, and Souleyrette (2012) analyzed VMT reductions using panel data for all states in the U.S. Kweon, and Kockelman (2004) analyzed VMT effect on household

characteristics using nation 1995 Nationwide Personal Transportation Survey data and found that non-parametric regression method was best used for analyzing VMT. Bhat and Nair (2000) developed a fractional split model that predicts the VMT mix on links as a function of the functional roadway classification of the link.

While VMT reduction policies have made their way into planning exercises and are tested in land use models, few of the reduction mechanisms are modeled within an empirically calibrated travel demand model or tested to determine how they would perform in a true environment examining user behavior to decision maker's policies. Further, a limited number of studies show application of travel behavior techniques to VMT reduction policies within a single modeling framework and compare the effectiveness of such policies. This paper works toward a more comprehensive methodology for VMT reduction policy evaluation. While travel demand models are used for a number of transportation planning and decision making purposes, it will be a worthwhile effort to incorporate strategies made state and county level to reduce VMT in travel demand modeling framework to envisage the traveler's response to such policies.

The objective of the paper is threefold. (1) Formulate models incorporating substantial strategies considered for VMT reduction. (2) Develop analytical framework to apply such models in a real world multimodal transportation network to estimate VMT reduction. (3) Compare the model results and derive conclusions on model performance. In the next section model formulation is presented.

# 3 FORMULATION

The model formulation section presents ways to incorporate specific strategies as a vision to reduce VMT and examine user's behavior using a travel demand model. A number of models are proposed considering their perspective implication in practice and enumerated as below.

- VMT based pricing: Model-1
- Gas tax: Model-2
- Commute efficiency: Model-3
- Pricing with variable demand
    - Variable Demand with VMT based pricing: Model-4(a)
    - Variable Demand with Gas tax: Model-4(b)
    - Variable Demand with Commute efficiency: Model-4(c)

In this section each model is empirically presented and discussed how it is analyzed in a trip based model. Specific emphasis is given to the traffic assignment step. However, the proposed methodology also encompasses destination choice and mode choice components. While the traffic assignment step is different for all the models, destination choice and mode choice remains same for all. The notations used are presented below followed by the formulation.

| Notation | | Explanation |
|---|---|---|
| $\hat{d}_{ij}^{\varphi}$ | : | Potential demand between *i-j* |
| $\Phi_a$ | : | VMT cap for link $a$ |
| $AOC_m^p$ | : | Auto operating cost for mode *m* and purpose *p* |
| $C_a$ | : | The capacity for link $a$ |
| $C_{jn}$ | : | Correction term to compensate for sampling error in model estimation |
| $DA_m^p$ | : | Drive access time for mode *m* and purpose *p* |
| $D_{ij}^{-1}$ | : | The inverse demand function associated with O-D pair *i-j* |
| $D_{ij}^k$ | : | Various distance terms (linear, log, squared, cubed and square root) |

| | | |
|---|---|---|
| $E_j^k$ | : | Employment of type *k* in zone j |
| $IVTT_m^p$ | : | In-vehicle travel time for mode *m* and purpose *p* |
| $IWTa1_m^p$ | : | Initial waiting time less than 7.5 minutes for mode *m* and purpose *p* |
| $IWTb_m^p$ | : | Initial waiting time greater than 7.5 minutes for mode *m* and purpose *p* |
| $L_{ij}$ | : | The mode choice logsum between zone pair *ij* |
| $NOT_m^p$ | : | Number of transfers for mode *m* and purpose *p* |
| $N_n^k$ | : | Person, household or production zone characteristics for trip *n* |
| $PC_m^p$ | : | Parking cost for mode *m* and purpose *p* |
| $S_j$ | : | Size variable for destination zone *j* |
| $TET_m^p$ | : | Terminal time for mode *m* and purpose *p* |
| $TF_m^p$ | : | Transit fare for mode *m* and purpose *p* |
| $U_{ijn}$ | : | The utility of choosing a trip attraction destination *j* for a trip *n* produced in zone *i* |
| $U_m^p$ | : | Utility function for auto and transit travel |
| $WT_m^p$ | : | Waiting time for mode *m* and purpose *p* |
| $Z_j^k$ | : | Attraction zone characteristics |
| $d_{ij}^{\varphi}(x_{ij})$ | : | New demand between O-D pair *i-j* |
| $f_{ij}^r$ | : | Flow on path *r*, connecting each Origin-Destination (O-D) pair (*i-j*) |
| $l_a$ | : | Distance for link $a$ |
| $q_{ij}$ | : | Demand between each Origin-Destination (O-D) pair (*i-j*) |
| $t_a$ | : | Travel time for link $a$ |
| $t_a(x_a)$ | : | Travel cost on link *a* as a function of flow |
| $u_a$ | : | User cost for link $a$ |
| $u_a^I(x_a)$ | : | Travel cost function for Model-1 which incorporates VMT pricing term $\theta_a$ |
| $u_a^{II}(x_a)$ | | Travel cost function for Model-2 which incorporates gas tax pricing term $\sigma$ |
| $u_{ij}^c$ | : | Least cost path between O-D pairs *i-j* |
| $u_{ij}$ | : | $=t_{ij}$ travel cost between origin *i* and destination *j* |
| $x_a$ | : | Flow for link $a$ |
| $d_{ij}$ | : | The number of commuter trips between *i* and *j* |
| $d_{ij}^o$ | : | Optimal number of commuters between *i-j* pair |
| $O_i$ | : | Total trips originated from zone *i* |
| $D_j$ | : | Total trips attracted to zone *j* |
| $\alpha_a$ | : | Constant, varying by facility type (BPR function) |
| $C_1, C_2$ | : | Commuter costs before and after optimization |
| $T$ | : | Total number of commuters |
| $\beta_a$ | : | Constant, varying by facility type (BPR function) |
| $\beta^k$ | : | Weights for each term in the size variable ($S_j$) |
| $\gamma^c$ | : | Value of time (VOT) for user class c |
| $\delta_{a,ij}^r$ | : | Flow on link $a$, a subset of path *r*, connecting each Origin-Destination (O-D) pair (*i-j*) |
| $\pi_m^p$ | : | Mode specific constant for mode *m* and purpose *p* |
| $\tau_a$ | : | Toll value for link $a$ |
| $\Phi_a$ | : | Total number of vehicles traversing link $a$ |
| $\theta_a$ | : | VMT based tax for link *a* |

| | | |
|---|---|---|
| $\sigma$ | : | Gas tax in \$/gallon |
| $\phi$ | : | Total VMT in the system |
| $y$ | : | Desired goal of VMT reduction |
| $c$ | : | User class |
| $t_o$ | : | Free flow time on link $a$ |
| $\beta$ | : | Mode and attribute specific coefficient (mode choice model) |
| $\vartheta$ | : | Automobile gasoline efficiency in miles per gallon |
| $\omega$ | : | A positive constant (exponential demand function) |
| $n$ | : | Assignment iteration number |

The fundamental aim of traffic assignment process is to reproduce on the transportation system, the pattern of vehicular trips/personal trips which would be observed when the travel demand (represented by the trip matrix, or matrices) is assigned to the network satisfied by employing behavioral models. The traffic assignment model is based on the behaviorally-sound principle of user equilibrium. This principle is based on the fact that individuals choose a route in order to minimize his/her travel time or travel cost and such a behavior on the individual level creates equilibrium at the system (or network) level over a long period of time (Sheffi 1984). Simply, for each origin-destination (O-D) demand pair, the travel-cost/travel-time on all used routes of the road network should be equal.

$$Minimize \sum_{a} \int_{0}^{x_a} (t_a(x_a)) \tag{1}$$

Subject to:

$$\sum_{r} f_{ij}^{r} = q_{ij} \tag{2}$$

$$x_a = \sum_{i} \sum_{j} \sum_{r} f_{ij}^{r} \, \delta_{a,ij}^{r} \tag{3}$$

$$f_{ij}^{r}, q_{ij}^{r} \geq 0 \tag{4}$$

Equation (1) represents that at equilibrium the network will satisfy User Equilibrium (UE) condition, i.e. travel time on all the used routes will be equal. The term, $t_a$, is the travel time for link *a*, which is a function of link flow $x_a$. Equation (2) is a flow conservation constraint to ensure that flow on all paths *r*, connecting each Origin-Destination (O-D) pair (*i-j*) is equal to the corresponding demand. In other words, all O-D trips must be assigned to the network. Equation (3) represents the definitional relationship of link flow from path flows. Equation (4) is a non-negativity constraint for flow and demand. The travel time function $t_a(.)$ is specific to a given link '*a*' and the most widely used model is the Bureau of Public Roads (BPR) function given by

$$t_a(x_a) = t_o\left(1 + \alpha_a \left(\frac{x_a}{C_a}\right)\right)^{\beta_a} \tag{5}$$

where *to*(.) is free flow time on link '*a*', and $\alpha_a$ and $\beta_a$ are constants (and vary by facility type). $C_a$ is the capacity for link *a*. In the base model the objective is explore user behavior till equilibrium is achieved.

### 3.1 First Best Vehicle Miles Travelled Pricing

The first best name comes from the idea that the pricing will charge all users for the average and marginal social cost of their externality. For example the resulting VMT in the current condition is $\phi$ and the goal is to reduce it by *y%*. The policy maker can create a threshold of desired VMT level of $\left(1 - \frac{y}{100}\right)\phi$. A threshold value for VMT based on total lane miles travelled can be determined on the network level, and each link can be subjected to be priced exceeding this threshold. The threshold value can be interpreted an additional charge per mile travel which represents the average cost to other drivers caused by driving is charged to each vehicle on the

network. Each driver is only charged for driving on a road that exceeds a pre-determined threshold and for the marginal amount of VMT. The threshold value can be considered as a cap. The cap can be determined by calculating the link level vehicle travel as the following.

$$\Phi_a = \left[\sum_{a=1}^{N}\left(\frac{\left(1-\frac{y}{100}\right)\phi}{n_a l_a}\right)\right] * n_a \tag{6}$$

where $\Phi_a$ is the total number of vehicles travelling for link $a$ calculated for each link in the base model, $l_a$ is the link distance, and $n_a$ is the number of lanes in link $a$. Once the cap is determined, the VMT based price ($\theta_a$) can be incorporated into the travel demand model. The emission price can be converted to travel time units with appropriate factor ($\frac{l_a}{\gamma^c}$) representing value of travel time (VOT) in monetary terms. The revised link based user cost function is

$$u_a^l(x_a) = t_a(x_a) + \frac{\theta_a l_a}{\gamma^c} \tag{7}$$

where $u_a^l(x_a)$ is the travel cost function for Model-1, which incorporates emission pricing term $\theta_a$. The objective function for Model-1 is similar to base case with the exception that the second term from equation (7) ($\frac{\theta_a l_a}{\gamma^c}$) is added to equation (1). The VMT price for each link is updated at the end of every assignment iteration based on the assignment produced as a result of that traffic flow at the iteration over the predetermined cap, so that changes in the results of each assignment are reflected in the travel cost faced by each user which like travel time, will vary between iterations.

The advantage of VMT based tax is to encourage travelers to use transit as an alternate mode if it the tax appears implausible. Equation (7) refers to VMT based tax associated with value of time (VOT). Users with a low VOT may consider using transit as an alternative mode

and vice versa for high income group travelers. From implementation viewpoint, fees could be collected annually through vehicle registration process, as mileage calculated through odometer readings. VMT based tax is intended as a price based dis-incentive to vehicular travel causing travelers to shift to other modes or make trips with shorter lengths resulting in lower VMT.

### 3.2 Gas Tax

A gas tax can be considered as another way of imposing a higher cost on highway travel. The gas tax differs from a VMT tax in three ways. First, the amount of gas consumed and thus the amount of taxes paid varies depending on the type of vehicle a road user drives, thus the driver has more control over total travel cost. Second, a gas tax is charged upfront (before a trip is taken) and generally hidden within the price of fuel, so users are less likely to link driving behavior to added fuel cost (Li et al., 2012). Third, while drivers do not closely link gas taxes to travel behavior like trip timing and route selection, studies have shown the drivers typically have a higher consumption elasticity for gas prices than for road charges, likely because of a difference in substitution options (Parry and Small, 2005). The effect of gas price on user behavior can be implemented as follows:

$$u_a^{II}(x_a, e_a,) = t_a(x_a) + \frac{\sigma l_a}{\gamma^c \vartheta} \tag{8}$$

where, $\sigma$ is the gas price in miles per gallon, $l_a$is the link length in miles, $\gamma^c$ is the VOT in \$/hr, and $\vartheta$ is the automobile gasoline efficiency in miles per gallon. Auto Operating Cost (AOC) is another component which is considered in the mode choice model (please see equation 22). A higher gas price will result in a higher AOC and therefore will make auto travel more expensive.

### 3.3 Commute Efficiency

The degree to which people live near their jobs is referred as commuting efficiency (Hamilton, 1982). The objective of commute efficiency is to facilitate more cohesive travel by altering the relative locations of jobs and housing. In contrast, excess commuting provides an objective measure of commuting efficiency given an observed or estimated commuting pattern. Excess commuting is therefore a rigorous benchmark for comparing commuting patterns among locations in a county or regional location. This policy determines the number of local jobs and the number of workers, then sets-up an optimization problem to determine the minimum distance workers would have to commute if as many local jobs were filled by local residence as possible (White 1988; Giuliano and Small 1993; Scott et al.1997; Niedzielski, 2006). Commute efficiency can be modeled as following.

$$Minimize \sum_i \sum_j u_{ij} d_{ij} \tag{9}$$

Where, $u_{ij} = t_{ij}$ , is the travel cost between origin $i$ and destination $j$ and $d_{ij}$ is the number of commuter trips between $i$ and $j$. The constraints for the optimization problem are

subject to

$$\sum_j d_{ij} = O_i \tag{10}$$

$$\sum_i d_{ij} = D_j \tag{11}$$

$$d_{ij} \geq 0 \tag{12}$$

The average commute cost from the optimization can be interpreted as

$$C_1 = \frac{1}{T} \sum_i \sum_j u_{ij} d_{ij}^o \tag{13}$$

Where, T is the total number of commuters, and $d_{ij}^{o}$ is the optimal number of commuters between *i-j* pair. Before optimization, the average commuter cost was:

$$C_2 = \frac{1}{T}\sum_i \sum_j u_{ij} d_{ij}^{o} \tag{14}$$

The excess commute can be defined as the percentage difference between cost before and after optimization. This can be represented as

$$C_{excess} = \frac{(C_2 - C_1) * 100}{C_1} \tag{15}$$

The excess commute can be considered as a dis-benefit from the planner's viewpoint. Commute efficiency suggests that if policies are created for residents to closely link home or job locations with each other to minimize the work commute, many of the negative externalities such as urban sprawl, congestion, pollution, and social costs can be minimized. Real world applications such as the Southern California Association of Governments (SCAG) efforts in growth-management plan, which facilitates jobs–housing balance, to reduce the number of vehicle miles traveled (VMT) by 35% with proportional reductions in automobile emissions (Giuliano, 1991).

### 3.4 Variable Demand and VMT

Model-1 (VMT based pricing) and Model-2 (Gas Tax), and Model-3 (Commute Efficiency) provide an understanding of shifting routes and modes. These models will be very helpful to analyze the reduction of VMT. However, all aforementioned models do not consider the variability of demand because of changes in network conditions as a result of different policies. Alternatively, the users are not elastic to the policies in earlier three models. Demand elasticity

can be incorporated into the Model-1 by introducing an inverse demand function in the objective function. Modeling variable demand completely changes the objective function. This can be represented as follows:

$$Minimize \sum_{a} \int_{0}^{x_a} \left( t_a(x_a) + \frac{\theta_a l_a}{\gamma^c} \right) - \sum_{ij} \int_{0}^{q_{ij}} D_{ij}^{-1}(x_a) \tag{16}$$

Equation (16) we name as Model-4(a) which denotes VMT pricing with variable demand. This formulation allows the decision maker to model the elasticity of user behavior. The constraints for the variable demand model remain the same as in the Base-case (see equations 2-4). The inverse demand function $D_{ij}^{-1}(.)$ is associated with O-D pair *i-j*. An exponential demand function is then used which is a function of potential demand and least cost user paths to determine the new demand $d_{ij}^{\varphi}(x_{ij})$ between O-D pairs is given by

$$d_{ij}^{\varphi}(x_{ij}) = \hat{d}_{ij}^{\varphi} \exp(-\omega * u_{ij}^{c}) \ \forall\, i, j, \tau \tag{17}$$

where $\hat{d}_{ij}^{\varphi}$ is the potential demand between *i-j*, $u_{ij}^{c}$ is the least cost path between O-D pairs *i-j* and $\omega$ is a positive constant. In this case the positive constant is .01 and is specified from the literature (Ukkusuri and Patil 2009). It represents the typical highway user's response to changes in travel time. The constant is low which means that users react to decreases in travel time, by taking more trips and increases in travel time by taking fewer trips, but do not drastically alter trip rates with changes in time. This new demand is then fed back into the highway assignment model. The users in the *i-j* O-D pair are now elastic to the cost of travel ($u_{ij}^{c}$). Alternatively, as the cost increases the willingness to travel decreases. Similarly gas tax and commute efficiency based on variable demand can be represented as Model-4(b), and Model-4(c) shown in equation 18 and 19 respectively.

$$Minimize \sum_{a} \int_{0}^{x_a} \left( t_a(x_a) + \frac{\sigma l_a}{\gamma^c \vartheta} \right) - \sum_{ij} \int_{0}^{q_{ij}} D_{ij}^{-1}(x_a) \tag{18}$$

$$Minimize \sum_{a} \int_{0}^{x_a} (t_a(x_a)) - \sum_{ij} \int_{0}^{q_{ij}} D_{ij}^{-1}(x_a) \tag{19}$$

The threshold value for this case can be at the discretion of the decision maker.

### 3.5 Choice of Destination and Mode

The base case and all models also work on the choice of destination and mode depending on the respective cost functions. Because of varying cost functions, the destination and mode chosen for different model will change. The following paragraph explains how destination choice and mode choice are incorporated in the model.

The utility $U_{ijn}$ of choosing a trip attraction destination *j* for a trip *n* produced in zone *i* is given by:

$$U_{ijn} = S_j + \alpha L_{ij} + \sum \beta^k D_{ij}^k + \sum \beta^k D_{ij}^k N_n^k + \sum \beta^k Z_j^k + C_{jn} \tag{20}$$

Where, $S_j$ is the size variable for destination zone *j*, $L_{ij}$ is the mode choice logsum between zone pair *ij*, $D_{ij}^k$ represents the various distance terms (linear, log, squared, cubed and square root), $N_n^k$ represents person, household or production zone characteristics for trip *n* and is used for creating interaction variables with distance terms, $Z_j^k$ represents attraction zone characteristics (other than the size term), and $C_{jn}$ is a correction term to compensate for the sampling error in the model estimation (i.e., it represents the difference between the sampling probability and final estimated probability for each alternative). The size variable may consist of several different terms; up to

four categories of employment in addition to households were used. Weights ($\beta^k$) for each term in the size variable were estimated along with all other model parameters as follows, where $E_j^k$ is employment of type *k* in zone *j*:

$$S_j = log\left(\sum \beta^k E_j^k\right) \tag{21}$$

The destination choice model provides O-D demand for all trip purposes. A nested logit structure is formulated for mode choice, which is based on generalized utility functions for auto and transit travel. Separate utilities were developed to represent mode choice by trip purpose. The mode choice utility function is represented as follows:

The complete utility function for mode choice is as follows:

$$\begin{aligned} U_m^p = \pi_m^p &+ \beta_{1_m}^p IVTT_m^p + \beta_{2_m}^p TET_m^p + \beta_{3_m}^p \times AOC_m^p + \beta_{4_m}^p PC_m^p + \beta_{5_m}^p \tau_m^p \\ &+ \beta_{6_m}^p WT_m^p + \beta_{7_m}^p IWTa1_m^p + \beta_{7_m}^p IWTb_m^p + \beta_{8_m}^p NOT_m^p \\ &+ \beta_{9_m}^p TF_m^p + \beta_{10_m}^p DA_m^p \end{aligned} \tag{22}$$

Where $\pi_m^p$ is a mode specific constant for mode *m*, and purpose *p*; $\beta$ in each term is the mode and attribute specific coefficient; $IVTT$ is the in-vehicle travel time, $TET$ is the terminal time, $AOC$ is the auto operating cost, $PC$ is the parking cost; $\tau$ is the toll value, $WT$ is the waiting time, $IWTa$ is the initial waiting time less than 7.5 minutes; $IWTb$ initial waiting time greater than 7.5 minutes; $NOT$ is the number of transfers, $TF$ is the transit fare; and $DA$ is the drive access time. The mode choice model results in splitting O-D trip matrices into 11 travel modes (3 auto modes, and 8 transit modes). Three auto modes refer to Single Occupant Vehicles (SOV), High Occupant Vehicles with 2 occupants (HOV-2), and High Occupant Vehicles with

three or more occupants (HOV-3+). Eight transit modes are walk and drive to bus, express bus, rail, and commuter rail.

# 4 SOLUTION METHODOLOGY

Figure 1 shows a flowchart of the solution algorithm for both the Base-case and four proposed models. For base case and Mode-1 through Model-3 the algorithm relies first on inputs commonly found in demand models, which includes the characteristics of the transportation networks (highway and transit), socio-economic and other inputs needed for the trip generation, destination choice, mode choice and traffic assignment programs. The traffic assignment is solved with a FW algorithm. Travel costs for Base-case are based on free flow travel times. For Model-1 and 2 the costs are based on equation 7 and 8 respectively. For Model-3 costs are based on free flow travel time but commute efficiency allows number of trips to find destinations close to the household location (please refer to equation 9 through 15).

For Model-4, variable demand is calculated based on the least cost path. The variable demand model first skims the Base-case assigned network for congested travel times. The shortest path between all O-D pairs is calculated and used in the inverse demand function to calculate the new demand based on the shortest path. Once the new demand is calculated it is used in the assignment model. The assignment model then iterates until the convergence criteria for the assignment program is met. Once the assignment convergence is met the demand is updated as per equation 17 for Model-4. The model iterates with variable demand until convergence. The complete model structure is incorporated into sequential scripts and finally called into a master batch file.

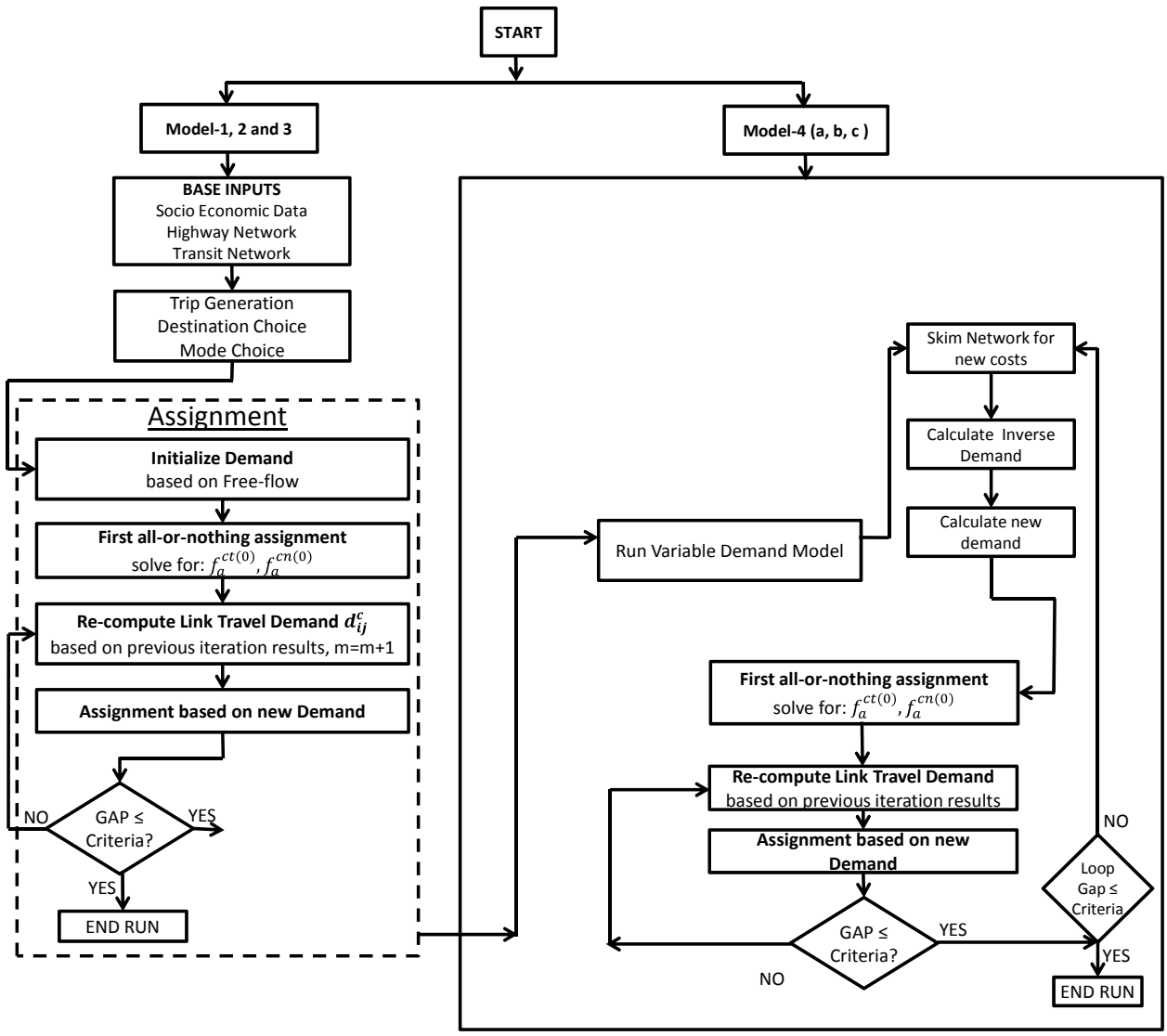


Figure 1: Solution Methodology

A step by step description of the assignment algorithm is presented below.

***Step 0: Initialization.***

Calculate initial demand ($q_{ij}^{c(0)}$) and feasible flow pattern ( $f_{ij}^{c(0)}$), based of free-flow travel time. Set ($n$) = 0.

***Step 1a: Update.***

$$\text{Set} \quad t_a(x_a) = t_a\left(1 + \alpha_a \left(\frac{x_a}{C_a}\right)\right)^{\beta_a} \quad \forall \quad a;$$

***Step 1b: Update with variable demand.***

Update demand with inverse demand function

$$d_{ij}^{\varphi}(\sigma_{ij}) = d_{ij}^{\tau} \exp\left(-\omega * u_{ij}^{c(m)}\right) \quad \forall_{ij,\tau}$$

where $u_{ij}^{c}$ is the least cost path between O-D pairs i and j and $\omega$ is a positive constant.

***Step 2: Direction Finding.***

Find the shortest path

$$u_{ij}^{c(n)} = t_a(x_a)$$

Perform all-or-nothing assignment based on updated travel times and obtain auxiliary flows $f_{ij}^{c(m)}$. Use the cost function shown in equation (7) and (8) for Model-1 and Model-2 and in Model- 4(a) and Model-4(b). Model-4(c) uses the cost function same as the Base-case.

***Step 3: Move Size.***

Line search for optimal step size, solving for *a*:

Base-case: $Minimize \ \sum_a \int_0^{x_a} (t_a(x_a))$

Model-1: $Minimize \ \sum_a \int_0^{x_a} \left(t_a(x_a) + \frac{\theta_a l_a}{\gamma^c}\right)$

Model-2: $Minimize \ \sum_a \int_0^{x_a} \left(t_a(x_a) + \frac{\sigma l_a}{\gamma^c \vartheta}\right)$

Model-3: $Minimize \ \sum_a \int_0^{x_a} (t_a(x_a))$ but with objective of commute efficiency

***Step 4: Flow Update with pricing scheme.***

*Find updated flow using travel cost functions for respective models.*

$$u_{ij}^{c(n)}(x_a) = t_a(x_a) + \frac{\theta_a l_a}{\gamma^c}$$

***Step 5: Assignment convergence criterion.***

$$K \leq \frac{Abs\left[\sum_{a\in A}\left(f_a^{(n)} * t_{a_a}^{c(n)}\right) - \sum_{a\in A}\left(f_a^{(n-1)} * \delta_a^{c(n-1)}\right)\right]}{\sum_{a\in A}\left(f_a^{(n-1)} * \delta_a^{c(n-1)}\right)}$$

where K is a dimensionless convergence criterion.

For Model-1, Model-2, and Model-3 step 1(b) is skipped. For Model-4, all the steps are executed. If inequality holds, terminate assignment and go to step 1a. Otherwise, set $n = n + 1$ and go to step 1.

# 5 DATA

The Montgomery county study area consists of 223 zones and 49,729 O-D pairs. The data is obtained from the Maryland Statewide Transportation Model (MSTM) developed by The Maryland State Highway Administration (SHA). Montgomery is the most populous county in the state with a population close to one million, 400,000 households, and employment of 600,000. The county boundary with the transportation network is presented in Figure 2. The county contains parts of the heavily travelled roadways in the Washington DC-Baltimore region (Washington DC is referred to as Washington in the remainder of the paper).

There are a total of 4,302 highway links in Montgomery County consisting of all facility types representing a total of 3,630 lane miles. In addition, the network consists of 73 bus lines. A number of transit options are available in the network. In terms of transit, the commuter rail MARC, which travels between Brunswick (North) to Union Station (South), travels completely through Montgomery County. A major metro line (called the Red Line) also travels through the county. In addition, Washington Metropolitan Transit Authority (WMTA) provides bus service and the county has extensive coverage of its own bus system called “RIDE-ON” with very high capacity and frequency.

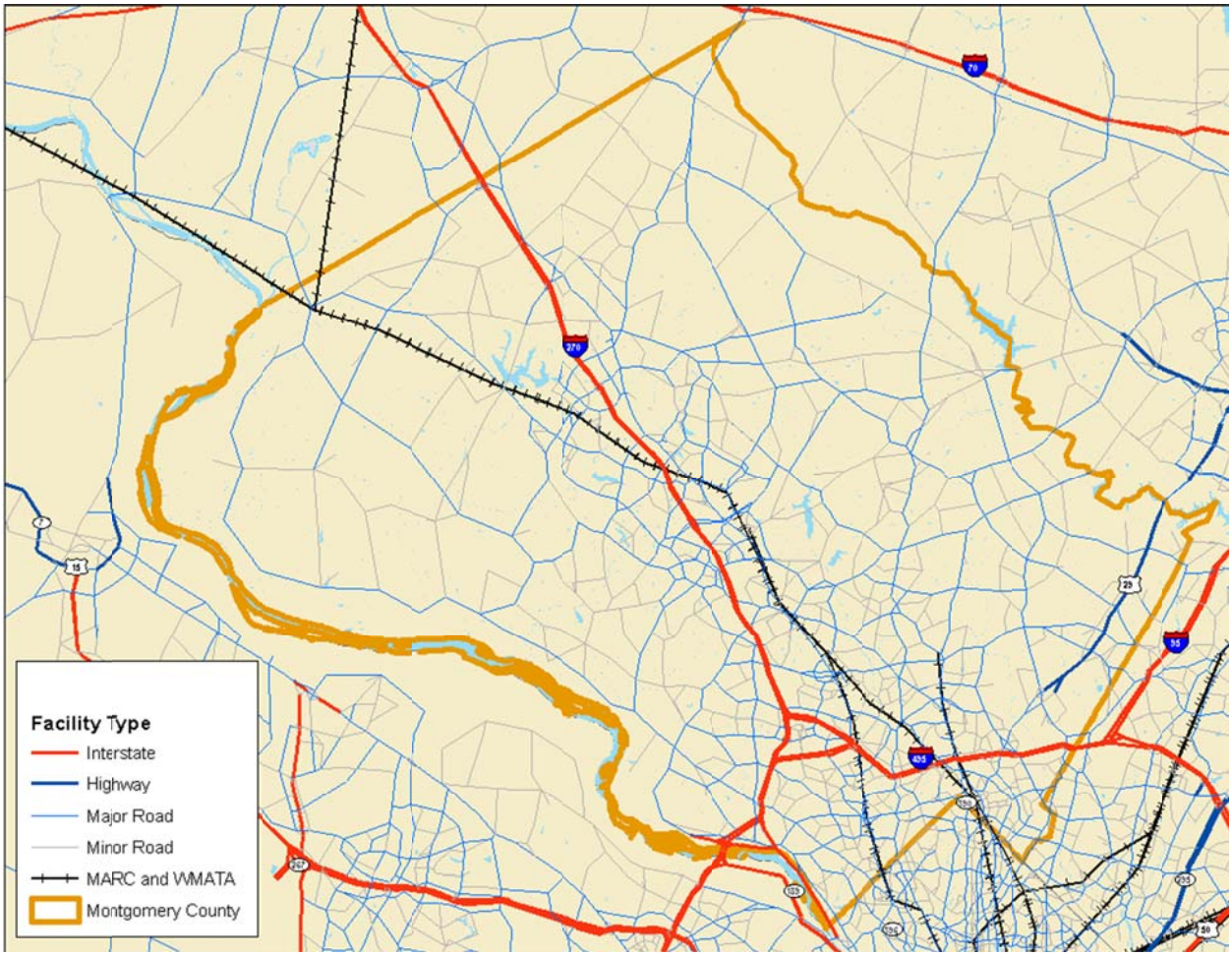


**FIGURE 2 Montgomery County Transportation Network**

The highway network, transit network, and trip generation data is obtained from SHA. Zonal, highway and transit network attributes (zonal household, employment, link distance, travel time, free flow speed, link capacity, transit stop location, direction of travel, frequency, bus /rail capacity) were borrowed from MSTM. The destination choice and mode choice parameters were calibrated and developed as part of MSTM from a recently conducted Household Travel Survey in the Baltimore-Washington region. Details of the parameter explanation are beyond scope of this paper. However, a complete documentation can be found in the user's guide (MSTM 2011). It should be noted that for this research none of the MSTM model components are used and completely different algorithms for assignment were developed.

For example, variable demand is not a component MSTM and the models developed were solely for research purposed and not been part of any reporting purposes before.

# 6 EMPIRICAL RESULTS

The empirical results section is organized in two parts. First part provides results from Model-1, Model-2, and Model-3. In the second part, results from Model-4(a), Model-4(b), and Model-4(c) is presented. The destination choice model adequately distributes the trips between O-D pairs as per the formulation presented in Equation (20). Further, the mode choice model splits trips between auto and transit. Additionally, the assignment process provides the shortest user cost paths to the O-D pair and represents user's response. Details of destination and mode choice are not presented in the paper for brevity and much emphasis is given to highway assignment.

## 6.1 Results from Model-1, 2, and 3

In the first part, a summary of the models is described and statistics including VMT, Congested lane Miles and average link speed for four types of highway facilities. Second, specific results from the excess commuting problem are presented. Third, link level performance by VMT and link capacity group is shown. Lastly, VMT profile across the network is shown for all model results.

Table 1 shows the VMT reduction resulting from each policy along with the congested lane miles and average link speed. Congested lane miles are the links that have a volume to capacity (V/C) ratio greater 0.75. In the Base-case VMT is a little more than 4.4 million for all facility types. Freeways produce the most VMT followed closely by major arterials. The base assignment results show that there are 218 congested lane miles, most of which is on major arterials.

**Table 1 Highway Assignment Results for Base-case and three Models**

| Model (1) | Facility Type (2) | VMT (3) | Congested Lane Miles (4) | Average Speed (MPH) (5) | Percent Change VMT (6) | Percent Change Congested Lane Miles (7) |
|---|---|---|---|---|---|---|
| **Base-case** | **Freeways** | 1,644,650 | 61 | 44.86 | | |
| | **Major Arterials** | 1,521,594 | 88 | 26.16 | | |
| | **Minor Arterials** | 510,272 | 35 | 25.46 | | |
| | **Other** | 770,421 | 34 | 19.33 | | |
| | ***TOTAL*** | ***4,446,938*** | ***218*** | ***29*** | | |
| **Model-1** (First Best) | **Freeways** | 1,398,256 | 44 | 47.25 | -14.98% | -26.76% |
| | **Major Arterials** | 1,345,898 | 54 | 26.64 | -11.55% | -38.45% |
| | **Minor Arterials** | 460,903 | 27 | 25.61 | -9.68% | -23.67% |
| | **Other** | 754,388 | 29 | 19.34 | -2.08% | -12.52% |
| | ***TOTAL*** | ***3,959,446*** | ***155*** | ***30*** | ***-10.96%*** | ***-28.80%*** |
| **Model-2** (Gas Tax) | **Freeways** | 1,577,535 | 59 | 45.86 | -4.08% | -2.96% |
| | **Major Arterials** | 1,432,207 | 70 | 26.46 | -5.87% | -20.84% |
| | **Minor Arterials** | 475,878 | 28 | 25.60 | -6.74% | -20.90% |
| | **Other** | 750,202 | 29 | 19.36 | -2.62% | -12.77% |
| | ***TOTAL*** | ***4,235,823*** | ***186*** | ***29*** | ***-4.75%*** | ***-14.64%*** |
| **Model-3** (Commute Efficiency) | **Freeways** | 1,611,054 | 60 | 45.03 | -2.04% | -0.26% |
| | **Major Arterials** | 1,391,490 | 69 | 26.39 | -8.55% | -21.42% |
| | **Minor Arterials** | 462,126 | 28 | 25.63 | -9.44% | -22.03% |
| | **Other** | 755,108 | 28 | 19.35 | -1.99% | -15.18% |
| | ***TOTAL*** | ***4,219,778*** | ***186*** | ***29*** | ***-5.11%*** | ***-14.68%*** |

Each of the three models reduces VMT, but the level of reduction and the impact on other highways assignment characteristics differs significantly. Model-1 implements a VMT Tax of 44.8 cents per mile on light duty vehicles and 89.6 cents per mile for tucks. This is consistent with a number of studies in the literature (Greene, 2011; and Walters 1961). Equation 7 shows the model implementation of this policy, where $\theta_a$ is considered as the VMT tax. The VMT charge, which is applied to travel for all road users for specific link usage, results in the second biggest decrease in VMT (10.96%). Drivers in Model-1 are not able to realize travel cost savings by simply taking less travelled routes, since road users are charged for every mile of driving

within the study area regardless of the facility type they use or the amount of VMT already on the facility. This results in a smaller VMT reduction but has a significant positive impact on congestion. In the case, the number of congested lane miles is reduced for all facilities including minor arterials, for a cumulative 29.54% reduction.

The use of a gas tax is less effective in reducing VMT. This is an expected result both in a modeling and real world context. In both contexts the gas tax differs from VMT charge in that it is an implicit cost of automobile ownership rather than an explicit cost of travel behavior. In other words, drivers are conscious of how their driving behavior will result in additional costs when they are directly charged. However, when the additional charge is buried in the price of gas, the payment is made in advance and there is no explicit implication of travel behavior on the cost, drivers are less likely to change their travel plans. In this case, VMT is reduced for all facility types, with an aggregate reduction of 4.75%. The small VMT reduction has a much larger impact on congested lane miles, achieving an 11.77% reduction.

Model-3 attempts an indirect method of VMT reduction by optimizing commute travel time (and distance) by reducing the time residents spend driving to work. This is done by finding a suitable job location with a minimum commute time. The result is a VMT reduction of 5.11%. Congested lane miles are also reduced by 11.71%.

Table 2 shows the impact of the excess commuting problem on travel time and VMT. The study area, Montgomery County, was divided into 223 zones and then analyzed based on zonal density. Density in this case is measured as the total households and employment per acre. The table shows that there are more jobs per resident worker in higher density areas. This means translates to two important impacts: 1) the higher the urban density, the greater the potential for VMT reductions through changes in commuting and 2) generally higher density areas have more

of an excess commute but low density areas have nearly as much wasteful commuting. Thus there is room for both VMT and efficiency improvements for all types of urbanized areas.

**Table 2 Excess commuting, by urban density**

| Urban Density | Jobs per | VMT | | | Commute type | | |
|---|---|---|---|---|---|---|---|
| (Units Per Acre) | Resident Worker | Model-4 | Base | Reduction (%) | Required (min) | Base (min) | Excess (%) |
| 0 - 2.5 | 0.32 | 1,413,689 | 1,470,246 | -3.85 | 10.91 | 18.99 | 42.54 |
| 2.5 - 5 | 0.43 | 923,758 | 976,431 | -5.39 | 8.22 | 14.37 | 42.80 |
| 5 - 7.5 | 0.63 | 579,479 | 611,691 | -5.27 | 6.05 | 14.31 | 57.71 |
| 7.5 - 10 | 0.99 | 558,983 | 597,125 | -6.39 | 6.96 | 12.25 | 43.18 |
| 10+ | 3.78 | 572,172 | 620,645 | -7.81 | 4.16 | 11.28 | 63.12 |
| County | 1 | 809,616 | 855,228 | -5.74 | 7.26 | 14.24 | 49.87 |

On average, workers residing within the study area, commute for 14.25 minutes in peak hour traffic. The excess commute results indicate that the average required commute, that is the minimum commute time after optimizing for excess commuting, is 7.26 minutes which means that on average 49.87% of the current commute is wasteful.

At the link level, each model has a unique impact on total VMT per link and by capacity. Figure 3 shows the results on VMT for links grouped by peak capacity. In the Base-case, VMT is generally smooth in growth within each capacity class, though more erratic on higher capacity links. The peak travel occurs on higher capacity links with significant VMT only on a few lower capacity links. With Model-1 the links are ordered in the same way as in the Base-case. Model-1 shows VMT in terms of the difference between links VMT from first-best pricing. The results show the significant impact link-level pricing has on VMT. Links with a capacity between 1,700 and 2,400 vehicles have reductions in VMT in the range of 4,000 to 6,000 with a high of nearly 10,000 vehicle miles. A few higher capacity links pick up additional travel.

Model-2 also results in VMT reductions with links in the vehicle capacity range between 1,700 and 2,400 having the greatest reduction, but generally lower than 2,000 vehicle miles.

Relative to Model-1, the VMT reduction on lower capacity links is minimal. Model-3 provides VMT reductions in smaller magnitudes on individual links, but results in reductions across link capacities. The model reduces emissions much more for links in the 400 to 1,600 vehicle range than Model-2 does.

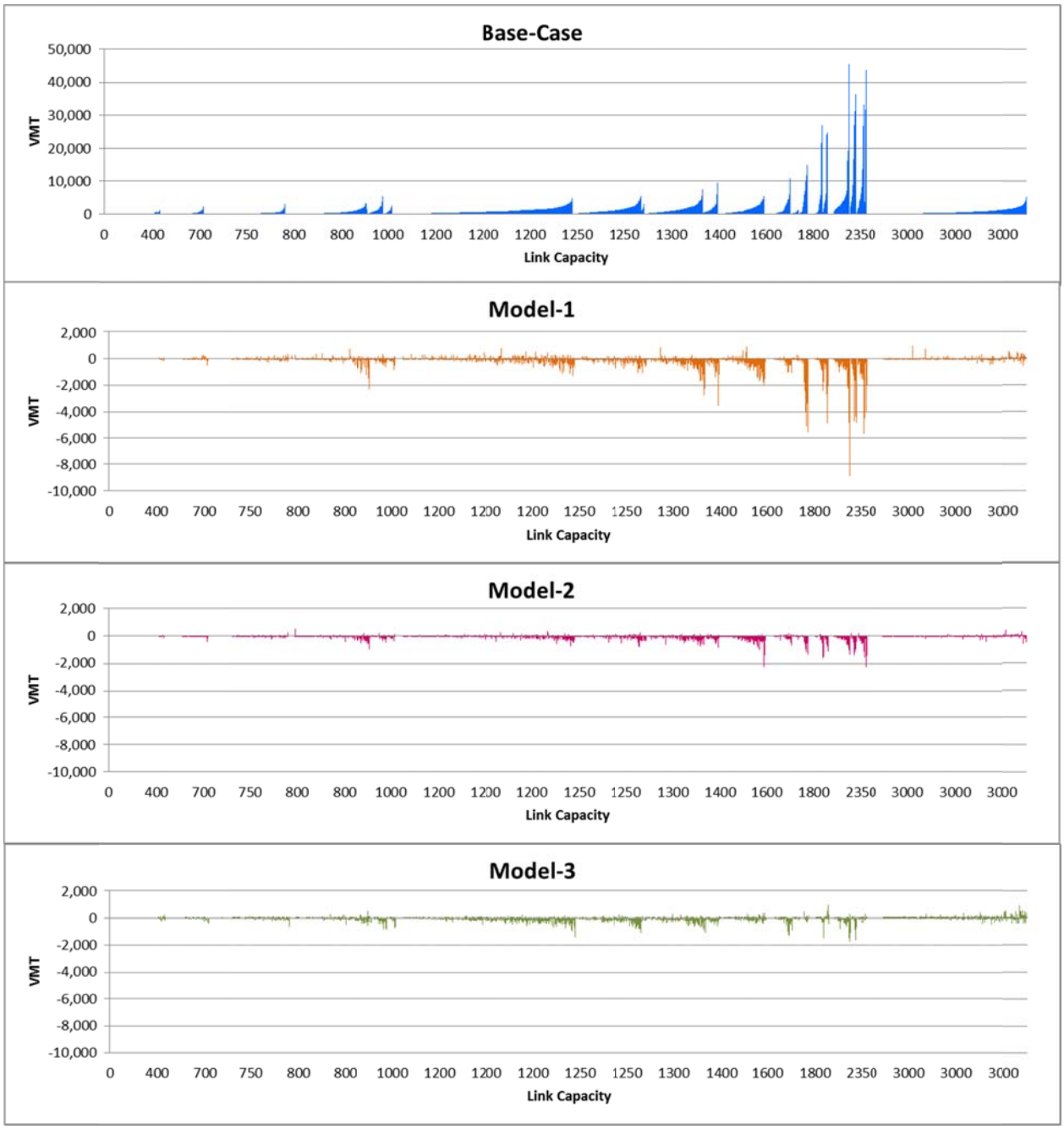


**Figure 3 Link VMT grouped by link capacity**

Figure 4 shows the effect of each model on VMT per capita (measured as total employment and households in a zone) based upon urban density. The results show that low density areas have high levels of VMT, which dramatically declines when density increases, which is consistent with finding in other studies (Chatman, 2003). Model-1 reduces VMT for all urban densities provides more of a reduction in VMT for less dense urban areas, Model-2 generally reduces per capita VMT for higher density areas (making more expensive to accumulate high VMT in rural areas), though in one case it does cause a slight increase. Model-3 produces the larges range of both reductions and increases in VMT for the lowest density areas. This is a result of the minimization in Model-3 of travel cost rather than distance, for some low density areas it becomes more cost effective to drive a longer distance.

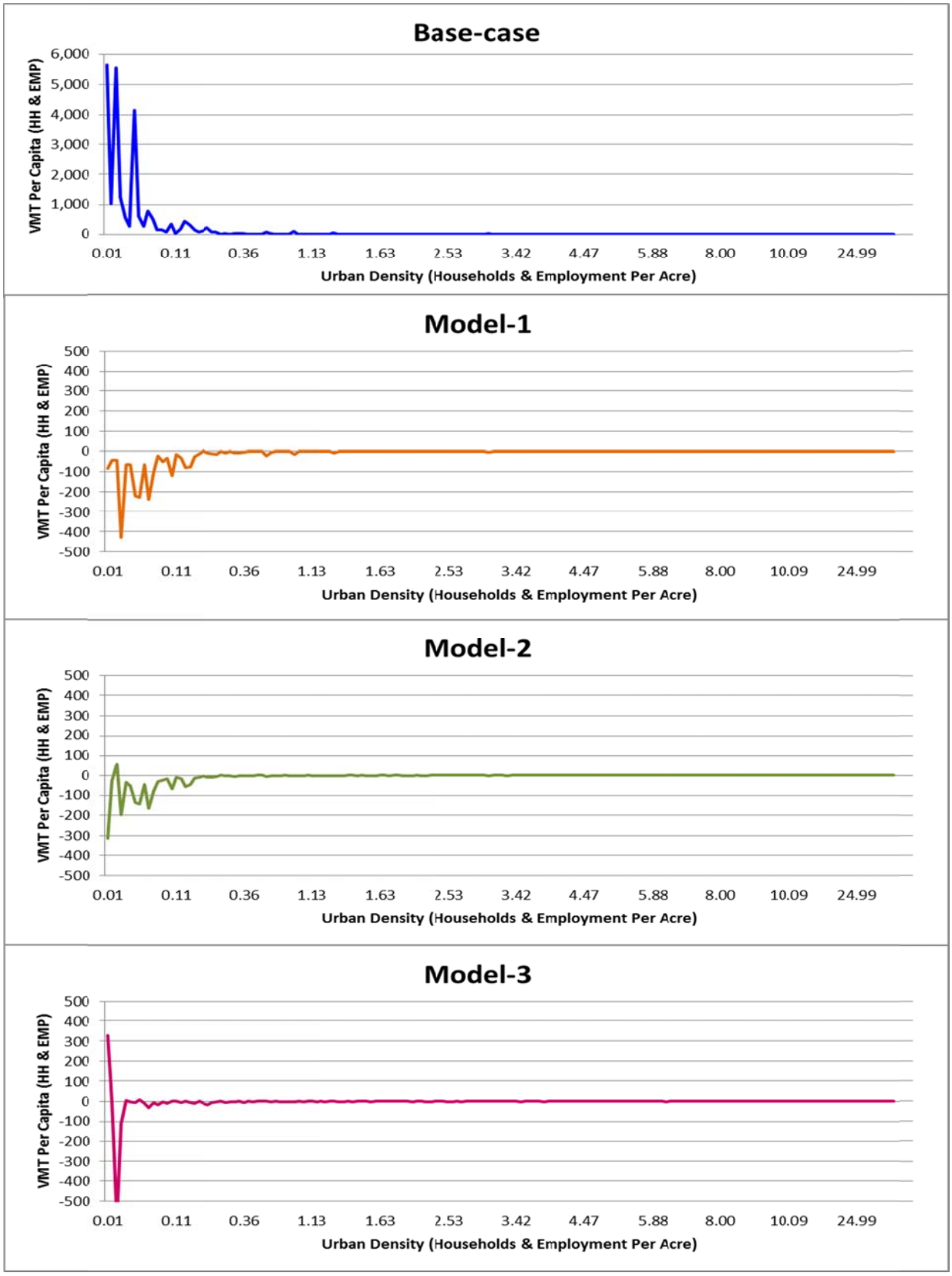


**Figure 4 VMT per capita by urban density**

Figure 5 shows the average relative per capita VMT change for each model. The results show that Model-1 produces the highest average VMT reductions for low density areas, while Model-3 has little impact on the same VMT. Model-2 results in a fairly stable VMT reduction of about 2% to 4% for most urban densities.

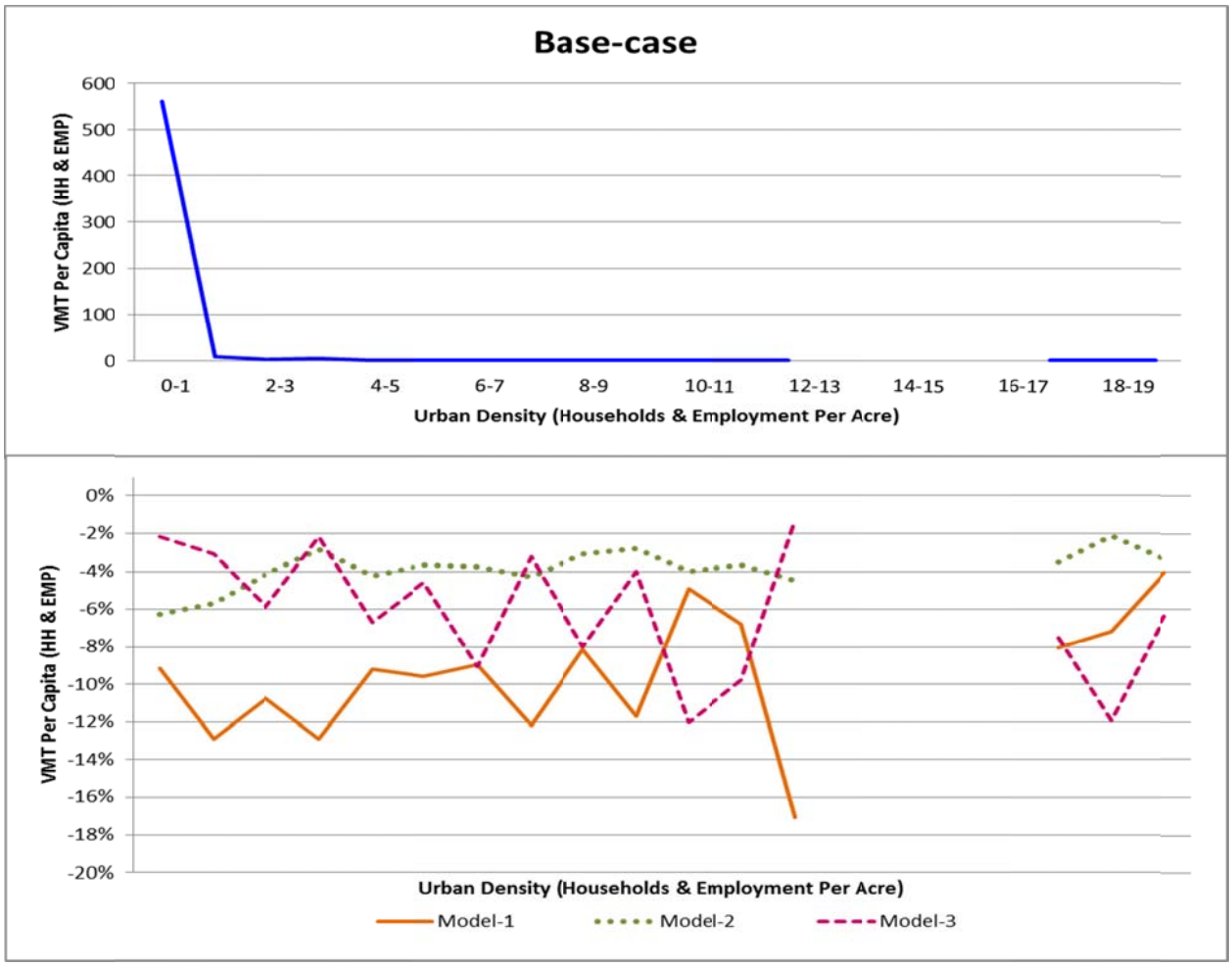


**Figure 5 Average VMT change per capita by urban density**

Figure 6 gives a three-dimensional representation of emissions for all four models and the Base-case, measured in VMT on each link. The result shows the magnitude and scale of VMT resulting trip assignment in each model.

In the first layer, above the highway network, is the Base-case. The level of VMT on the highways is apparent from the peaks that follow the interstates (the lower level). The second

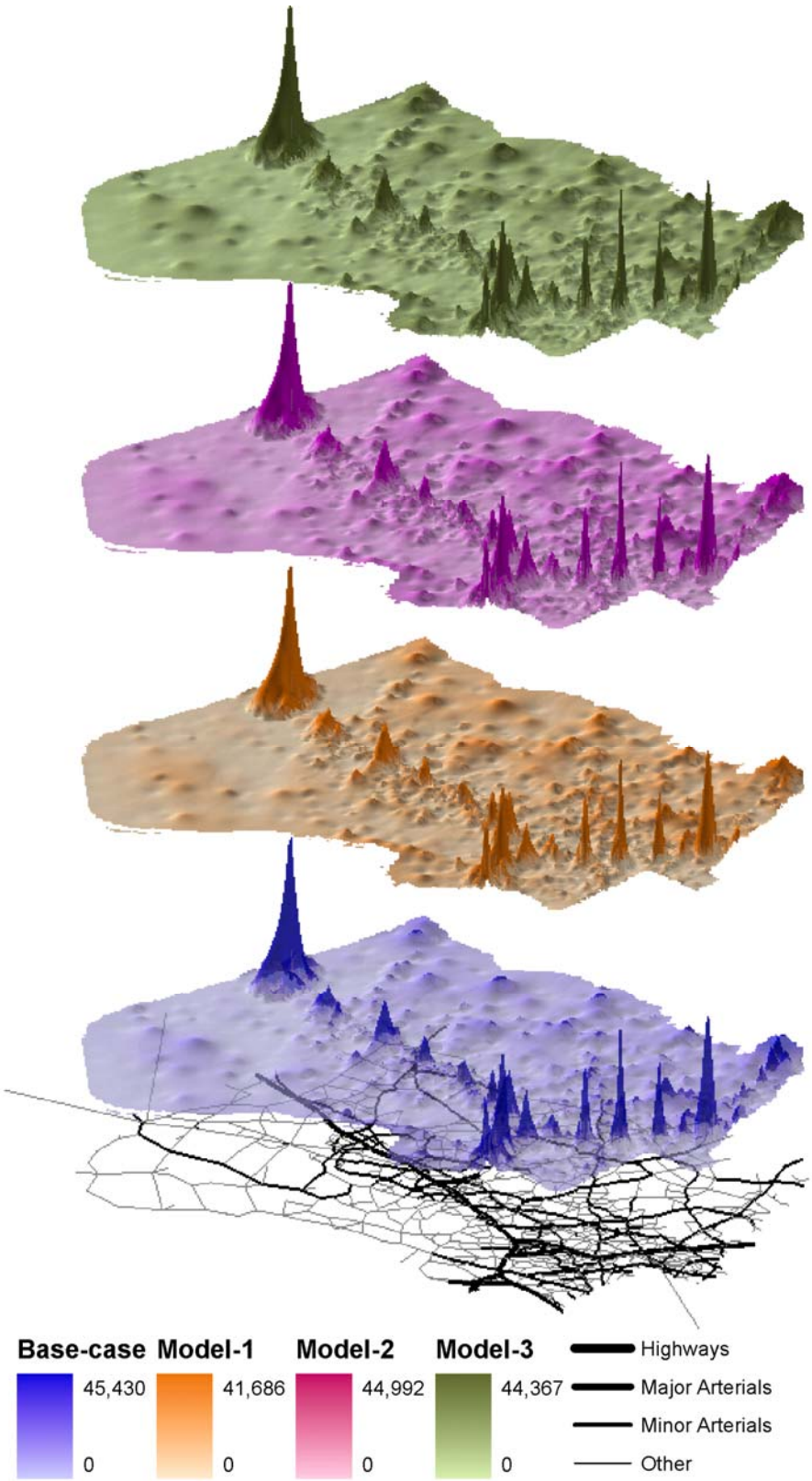


**Fig. 6 VMT Surface for Model-1, Model-2, and Model-3**

layer, a representation of VMT from Model-1 shows the effects of a direct VMT charge. The peaks away from the highway surface raise as some trips are diverted from the highway. However the average trend is similar to the Base-case. This is a result of the limited VMT tax avoiding behavior discussed above.

The final two models show the modest improvements over the Base-case. The magnitudes of VMT along certain segments of the highway are similar to the Base-case, but some of the VMT reductions are due to less highway driving and slightly more dispersed local road use. This is consistent with the two methodologies which both encourage less long distance travel and fewer internal-external and external-internal trips.

### 6.2 Results from Model-4(a), Model-4(b), and Model-4(c)

The format of 6.2 follows 6.1, but with an analysis of the models under a variable demand framework; first part provides a summary of the models including VMT, Congested lane Miles and average link speed for four types of highway facilities. Second, link level performance by VMT and link capacity group is shown. Lastly, link level VMT is shown for all model results.

When variable demand is added to the modeling framework, VMT is reduced even further. These reductions represent driver's decisions to not take a trip, modeling the elastic nature of travelers' behavior to the given network conditions. When pricing or strategy is employed in the variable demand model a number of users may first decide whether or not to make a trip subject to imposed user costs. The "willingness to travel" was not analyzed in the earlier set of static demand models. For the variable demand model, in equation (17) the value of $\omega$ is 0.01 representing the elasticity of users response to travel cost.

Table 3 shows the assignment results from the models with variable demand. Model-4(a) results in the highest VMT reduction. This is due to the added cost of travel from road pricing.

User response to pricing reduces VMT from 10.96% in the static demand model to 18.50%. The largest reductions in VMT occur on the lower facility types: minor arterials and other. Model-4(b) reduces VMT slightly more than Model-1, resulting in a 7.83% decrease over the static model, from 4.75% (Table 1) to 12.58% (Table 3). Model-4(c) results in the largest change in VMT from static demand conditions. The model produces and additional 10.06% reduction in VMT. This occurs to the way travels first responded to commuting efficiency, by utilizing more local roads when travel time was optimized. When the cost of travel then becomes a consideration in trip-taking decisions, the local roads see substantial VMT reductions.

**Table 3 Highway Assignment Results with Variable Demand**

| | | | | | Percent Change | |
|---|---|---|---|---|---|---|
| **Model (1)** | **Facility Type (2)** | **VMT (3)** | **Congested Lane Miles (4)** | **Average Speed (MPH) (5)** | **VMT (6)** | **Congested Lane Miles (7)** |
| **Base-case** | **Freeways** | 1,644,650 | 61 | 44.86 | | |
| | **Major Arterials** | 1,521,594 | 88 | 26.16 | | |
| | **Minor Arterials** | 510,272 | 35 | 25.46 | | |
| | **Other** | 770,421 | 34 | 19.33 | | |
| | ***TOTAL*** | ***4,446,938*** | ***218*** | ***29*** | | |
| **Model-4a** (First Best) | **Freeways** | 1,288,827 | 35 | 48.37 | -21.64% | -42.79% |
| | **Major Arterials** | 1,259,951 | 39 | 26.88 | -17.20% | -55.93% |
| | **Minor Arterials** | 422,276 | 21 | 25.74 | -17.25% | -41.92% |
| | **Other** | 653,338 | 26 | 19.36 | -15.20% | -22.79% |
| | ***TOTAL*** | ***3,624,393*** | ***120*** | ***30*** | ***-18.50%*** | ***-44.90%*** |
| **Model-4b** (Gas Tax) | **Freeways** | 1,472,652 | 50 | 47.41 | -10.46% | -16.69% |
| | **Major Arterials** | 1,333,865 | 51 | 26.73 | -12.34% | -42.00% |
| | **Minor Arterials** | 431,830 | 22 | 25.75 | -15.37% | -36.43% |
| | **Other** | 649,377 | 23 | 19.37 | -15.71% | -31.93% |
| | ***TOTAL*** | ***3,887,724*** | ***147*** | ***30*** | ***-12.58%*** | ***-32.51%*** |
| **Model-4c** (Commute Efficiency) | **Freeways** | 1,489,359 | 52 | 46.85 | -9.44% | -14.53% |
| | **Major Arterials** | 1,249,415 | 50 | 26.78 | -17.89% | -42.91% |
| | **Minor Arterials** | 405,824 | 19 | 25.82 | -20.47% | -46.27% |
| | **Other** | 627,949 | 22 | 19.38 | -18.49% | -34.52% |
| | ***TOTAL*** | ***3,772,547*** | ***143*** | ***30*** | ***-15.17%*** | ***-34.28%*** |

In terms of congested lane miles, Model-4(a) results in 44.9% reduction, followed by Model-4(c) and Model-4(b). Irrespective of VMT, congested lane miles, Model-4(a) produces highest reduction compared to its counterparts. In Figure 7, VMT reduction by link capacity for Model-4(a) through (c) is presented, along with the Base-case. Highest reduction in VMT is occurs in

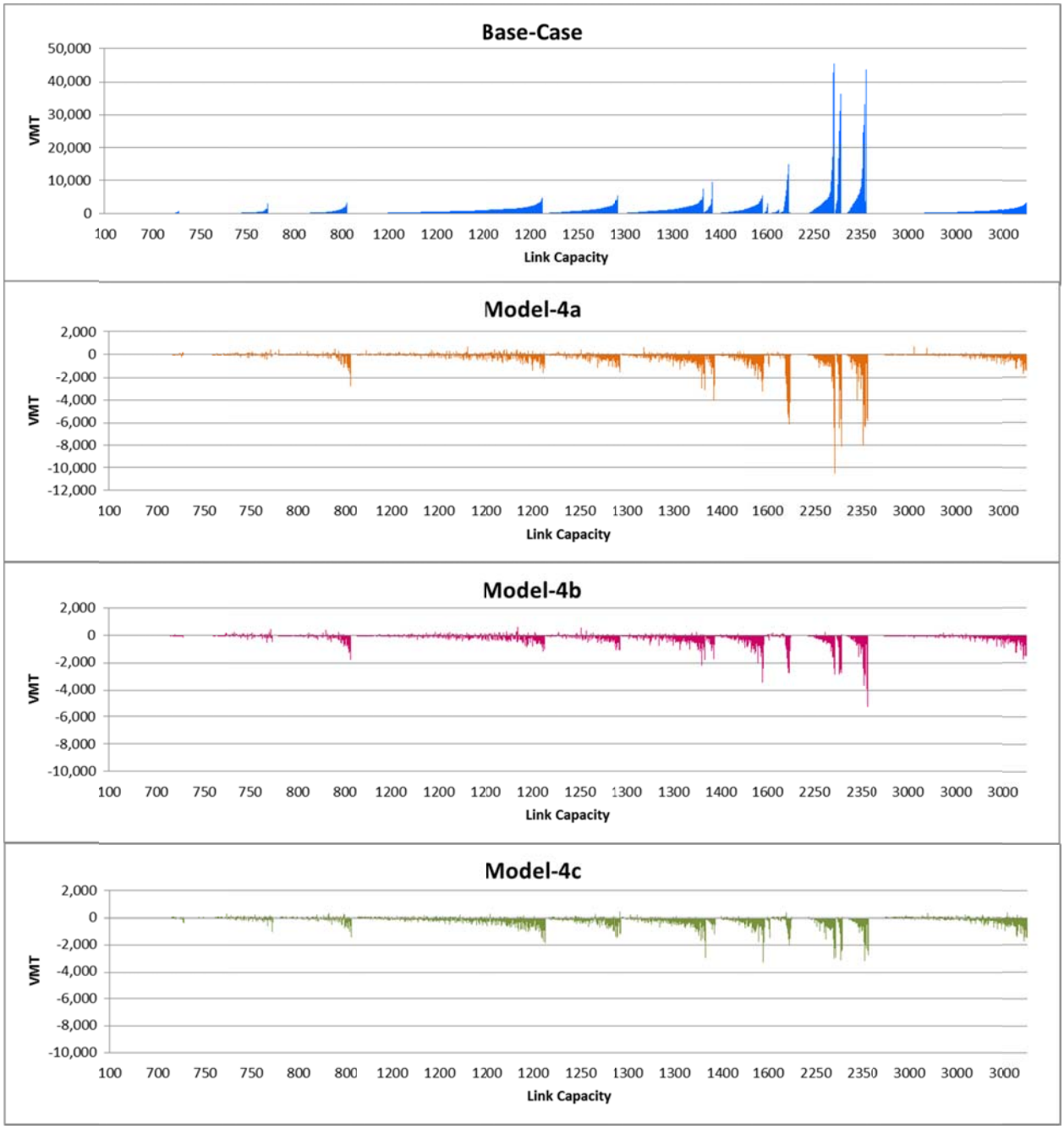


**Figure 7 Link VMT grouped by link capacity with Variable Demand**

the higher link capacities of Model-4(a). Lower reductions are observed for Model-4(b) and Model-4(c).

Figure 8 shows that Model-4(a) reduces VMT for all urban densities provides more of a reduction in VMT for less dense urban areas, Model-4(b) reduces per capita VMT for higher density areas (making more expensive to accumulate high VMT in rural areas), though in one case it does cause a slight increase. Model-4(c) produces the largest range of both reductions and increases in VMT for the lowest density areas. This is a result of the minimization in Model-4(c) of travel cost rather than distance, for some low density areas it becomes more cost effective to drive a longer distance.

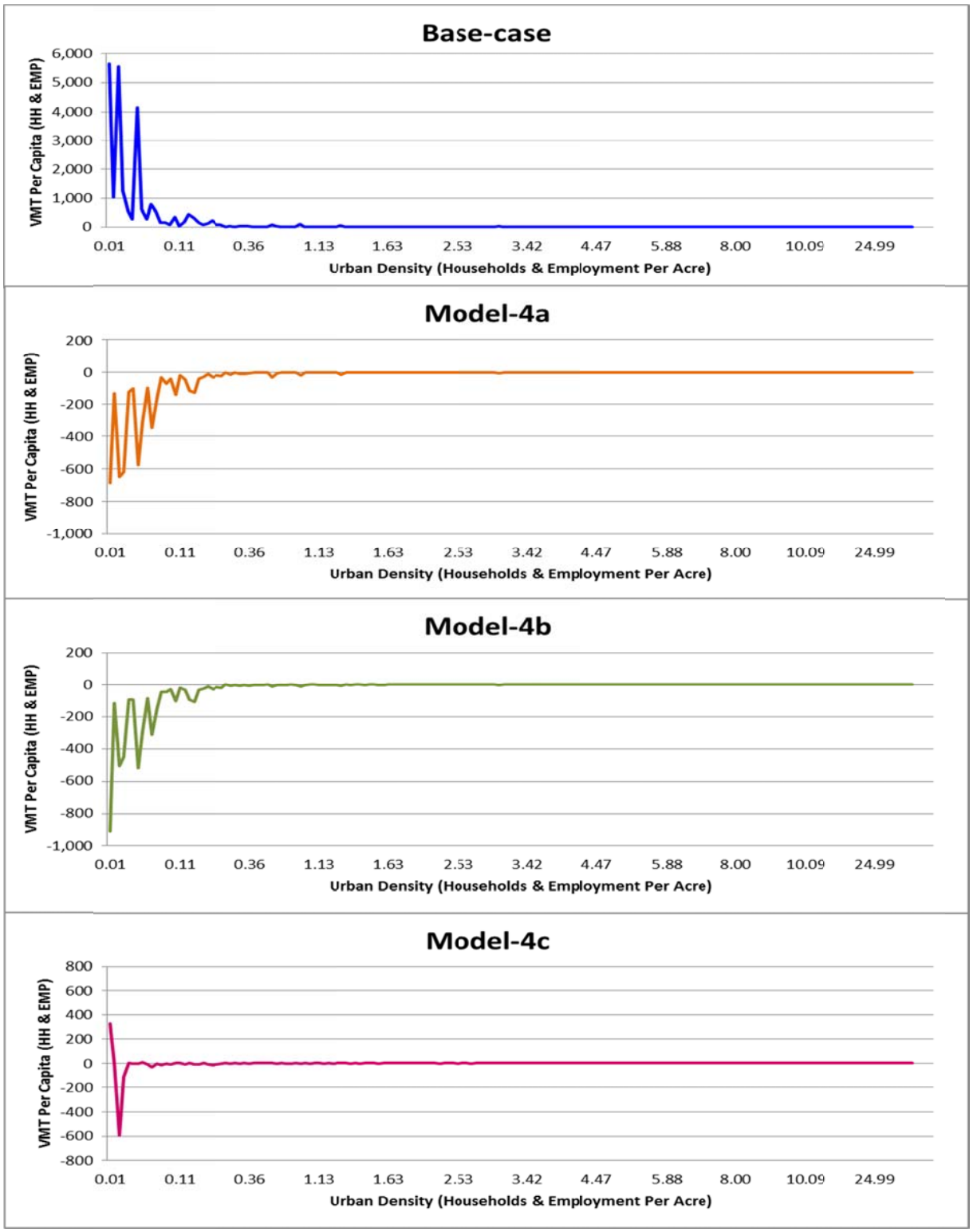


**Figure 8 VMT per capita by urban density with Variable Demand**

Figure 9 shows the average relative per capita VMT change for each model. The results show that Model-4(a) produces the highest average VMT reductions for low density areas, while Model-4(c) has little impact on the same VMT. Model-4(b) results in VMT reduction between its counterparts.

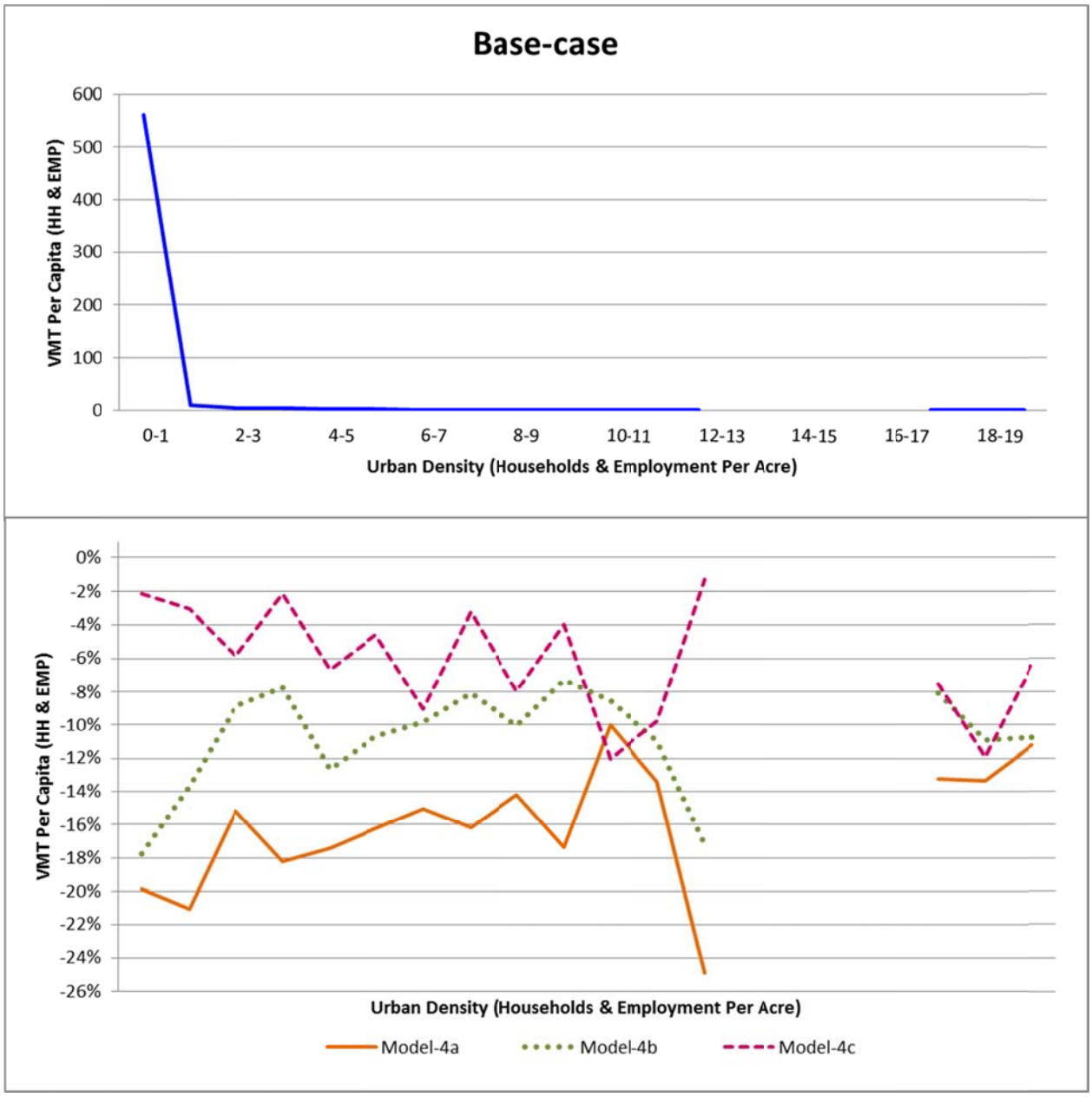


**Figure 9 Average VMT change per capita by urban density with Variable Demand**

In Figure 10, similar to Figure 6, the three dimensional diagram of VMT shows that Model-4(a) has largest VMT reduction in all facility types. Whereas other models show VMT reduction on

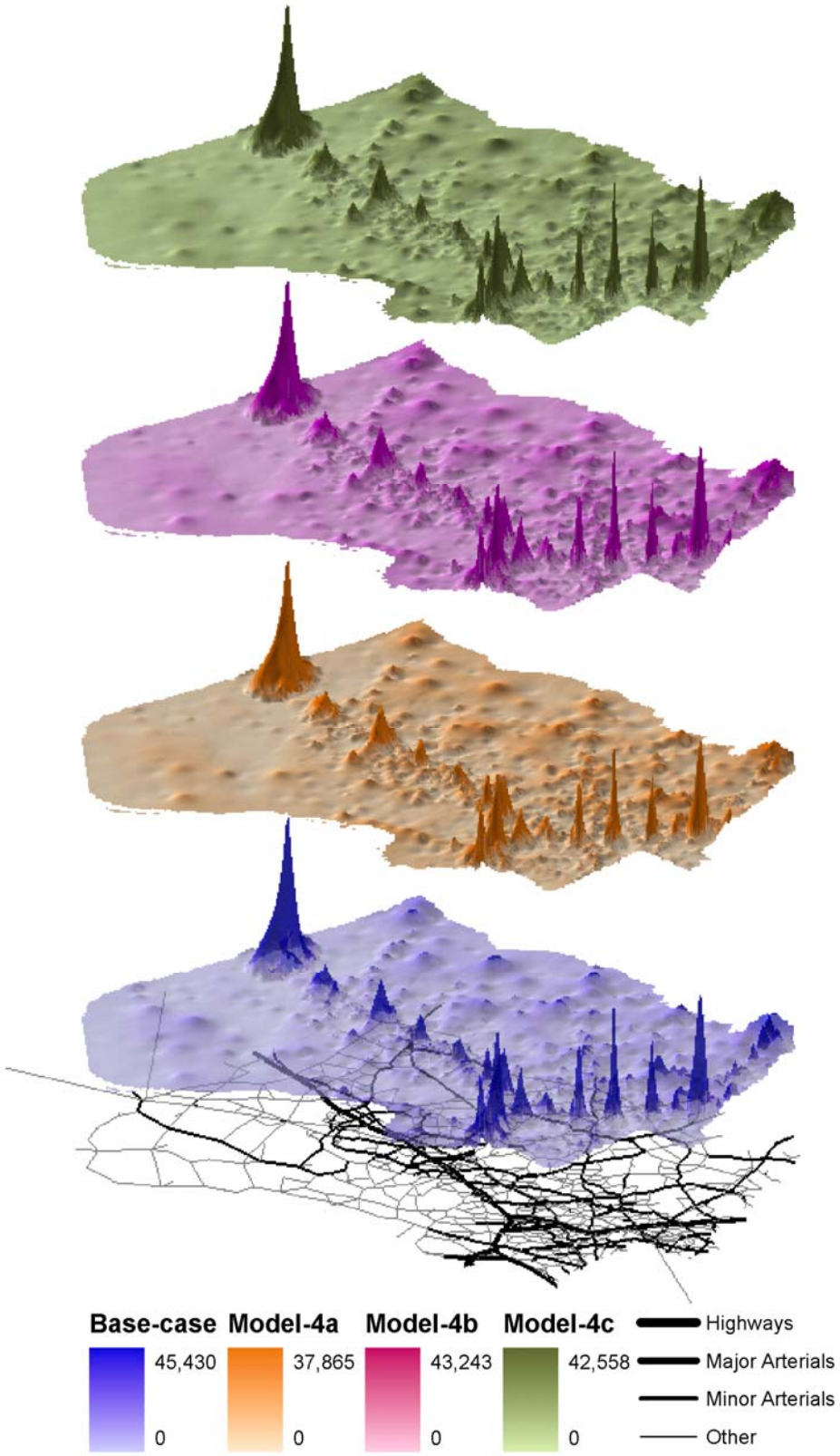


**Figure 10 VMT Surface for Model-4(a), Model-4(b), and Model-4(c)**

freeways, but the spread effect of VMT is shown in arterials and minor roadways. The reduction in VMT for Model-4(a) is largest among its counterparts and this effect is shown in the contour band in Figure 10. The four models presented had computational times per iteration of 12 minutes for Base-case and 16 minutes Model-1 through Model-3. Model-4's computation time was longer because of each variable demand iteration has a nested traffic assignment program that also had to iterate and converge. The Base-case and all models models were run in batch mode and scripts written in Cube Voyager, R and Visual Basic using eight-core distributed processing in a Windows Server 2008. However the processing times may vary as the computer configuration is altered and a different size of the problem is analyzed.

## 7 CONCLUSION AND RECOMMENDATIONS

In this paper travel demand formulation and application is proposed with vision to quantify VMT reduction strategies. The objective of the proposed travel demand model framework is to implement a set of strategies created by the decision maker and to assess the traveler's behavior in response to those strategies. As a part of formulation, a Base-case, and three models are proposed: (1) VMT based pricing, (2) gas tax, (3) commute efficiency. The Base-case portrays current travel conditions and three models incorporated policies under the assumption of trip based travel demand. Further these three models are analyzed assuming if the users are elastic to travel if the network conditions changes.

For the travel demand model zonal production and attraction data was *apiori*. In the formulation, destination choice, mode choice and trip assignment are discussed. For the Base-case and each model, different cost functions are derived. The cost function essentially

represents the policy maker's strategy for VMT reduction. To determine road user's response, the cost functions are used in the destination choice, mode choice and trip assignment procedure. Trip assignment results determine route choice behavior of users and provide link level traffic volume, congested travel time and speed levels.

The Base-case and each model is independently examined in a case study and also measured against each other, in a single application, to determine the most likely candidate to achieve substantial VMT reductions. The results of the models were compared for their effectiveness in the study area of Montgomery County, Maryland. Model results suggest that there is no single best strategy for VMT reductions.

For first three models it is seen that VMT based tax produced highest reductions compared to the Base-case. Highest reductions are observed on the freeways. In terms of congested lane miles also VMT based tax resulted in largest reduction. When the model performance is compared against urban densities VMT based tax shows higher reduction in low density areas, where gas tax shows low reduction in high density area allowing users to shift to transit mode. Commute efficiency model results show that VMT can be potentially reduced if employment and household locations are closely located. In terms of variable demand model, VMT based tax also demonstrates highest reduction in VMT.

The contribution of paper has several dimensions in terms of model formulation, application and contribution to practice. First, development of functional forms for strategic VMT reduction programs. Second, procedural application of four models that incorporate decision maker's strategies in VMT to examine user's response. Third, application of proposed methodology in a real world multimodal transportation network for comparison of model results and for quantification of benefits of each model.

This paper shows that achieving VMT reductions is a complicated but achievable goal, as there is an uncertainty of road user's response to each policy. The implementation of a single policy may have detrimental impacts either to other highway network conditions. The research leaves open the question of how large a policy area is needed and the level of regional coordination required to achieve optimal VMT reductions. In this paper trip based model is used to obtain effects of VMT reduction strategies while each individual characteristic are aggregated to zonal levels. To obtain each individual's trip making behavior and elasticity to such policy changes a micro level land use and tour based travel demand models options can be explored.

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